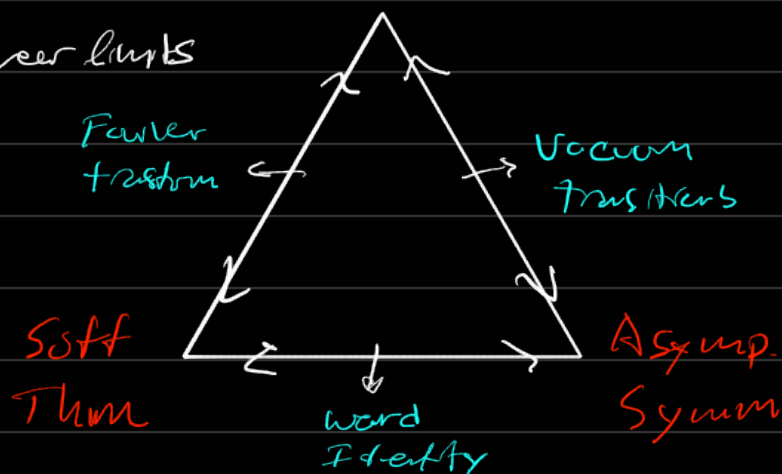


IR triangle:

Memory Effects

* IR Dynamics: soft/collinear links



* Soft thm: tell us that a large (infinite) # of soft particles are produced in any physical process but in a highly controlled manner.

* Memory effects: DC effect, in which the passage of gravitational waves produce a permanent shift in the relative positions of a pair of inertial detectors.

↳ Soft thm: statement about momentum space poles in scattering amplitudes

↳ Memory: DC shift in asymptotic data between late and early times.

⇒ F.T. of a pole in momentum space is a step fn:

$$\frac{d}{dt} \int d\omega \frac{e^{i\omega t}}{\omega} = \frac{d}{dt} F(\hat{p})(t)$$

$$i \int d\omega e^{i\omega t} = \frac{d}{dt} F(\hat{p})(t)$$

$$i \delta(t) \text{ but } \delta(t) = \frac{d}{dt} \Theta(t)$$

$$i \frac{d}{dt} \Theta(t) = \frac{d}{dt} F(\hat{p})(t) \quad \text{or}$$

$$i \Theta(t) = F(\hat{p})(t) \quad \text{as claimed.}$$

• $\Theta(t)$ can be understood as a domain wall connecting two inequivalent vacua that are related by an asymptotic symmetry (arxiv: 1411.5745)

* Just as in the vacuum state, if subleading momenta

and vacuum bubbles we have soft photons and gravitons which are produced.

→ QFT includes IR regulators which break the symmetry.

QED: (Lienard-Wiechert Solution)

$$S = -\frac{1}{4e^2} \int d^4x \sqrt{-g} F_{\mu\nu} F^{\mu\nu} + S_M, \quad U(1)$$

$$F^{(2)} = dA^{(1)} \quad \hookrightarrow \int *F \wedge F + S_M$$

$$j^\nu(x) \equiv -\delta S_M / \delta A_\nu(x)$$

$$\frac{\delta S}{\delta A_\nu} = \partial_\mu (\sqrt{-g} F^{\mu\nu}) + e^2 \sqrt{-g} j^\nu = 0$$

Note $(*F)_{\mu\nu} = \frac{1}{2} \sqrt{-g} \epsilon_{\mu\nu\rho\sigma} F^{\rho\sigma}$
 and $*F^{\mu\nu} = \frac{1}{2} \frac{1}{\sqrt{-g}} \epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}$

→

$$\boxed{d * F = e^2 * j} \Rightarrow \nabla^\mu F_{\mu\nu} = e^2 j_\nu$$

• $(*d * F)_\nu = \partial_\mu F^\mu{}_\nu \quad | \partial \Sigma_3 = \Sigma_2$

$$\Rightarrow Q_E = \frac{1}{e^2} \int_{\Sigma_3} d^3x \sqrt{-g} j^0 = \int d^3x \partial_i (\sqrt{-g} F^{0i})$$

$$Q_E = \frac{1}{e^2} \int_{\Sigma_2} *F = \int_{\Sigma_2} *j \in \mathbb{Z}$$

Full details of the above:

$$(*j)_{\mu_1\mu_2\mu_3} = e \epsilon_{\mu_1\mu_2\mu_3\rho} j^\rho g^{\sigma\rho} = \sqrt{-g} \epsilon_{\mu_1\mu_2\mu_3\rho} j^\rho$$

$$\hookrightarrow \int *j = \int \epsilon_{123\rho} j^\rho \sqrt{-g} dx^1 dx^2 dx^3$$

and so $\rho = 0$

$$= \int d^3x \sqrt{-g} j^0$$

$$\text{and } \int_{\Sigma_2} *j = \frac{1}{e^2} \int_{\Sigma_2} d * F = \frac{1}{e^2} \int_{\Sigma_2} *F = \frac{1}{e^2} \int_{\Sigma_2} *F$$

* $Q_E \in \mathbb{Z}$ (imposed)

electron: $Q_E = -1$,

$$Q_M = \frac{1}{2\pi} \int F$$

• Gauge: $\delta_\varepsilon A = d\varepsilon$, $\delta_\varepsilon \Phi_k = i\varepsilon Q_k \Phi_k$

$$\varepsilon \sim \varepsilon + 2\pi$$

$$Q_k \in \mathbb{Z}$$

$$\Phi_k \rightarrow e^{iQ_k \varepsilon} \Phi_k$$

• Consider a source j corresponding to n -particles each moving with constant velocity

$$U_k^\mu = \gamma_k (1, \vec{B}_k) \quad U_k^2 = -1 \quad \gamma_k = \frac{1}{1 - \vec{B}_k^2}$$

$$\Rightarrow j_\mu(x) \sim Q/\text{vol}, \quad \delta^{(n)} \sim [x]^n$$

• Each particle has position

$$x^\nu = U_k^\nu \tau$$

$$U_k = \frac{dx^\mu}{d\tau} = \left(\frac{dt}{d\tau}, \frac{dx^i}{d\tau} \right) = \left(\frac{dt}{d\tau} \right) \left(1, \frac{dx^i}{dt} \right)$$

$$\frac{dt}{d\tau} = \gamma_k, \quad \frac{dx^i}{d\tau} = B_k^i$$

$$U_k^\mu = \frac{dt}{d\tau} \left(1, \frac{dx^i}{d\tau} \right)$$

$$U_k^\mu U_{k\mu} = \left(\frac{dt}{d\tau} \right)^2 (-1 + B_k^2) = -1$$

• Set $x^\mu(0) = 0$, $x_k^\mu(t) = U_k^\mu \tau$

$$j^\mu(x) = \int d\tau \delta^{(4)}(x^\mu - U_k^\mu \tau)$$

$$\Delta j \sim \rho Q : \quad \rho Q = \sum_{n=1}^n Q_n \int d\tau \frac{dx^\mu}{d\tau} \delta^{(4)}(x^\mu - U_n^\mu \tau)$$

so

$$j_\mu(x) = \sum_{n=1}^n Q_n \int d\tau U_{n\mu} \delta^{(4)}(x^\mu - U_n^\mu \tau)$$

Solution: $\frac{e^2}{4\pi} \sum_{n=1}^n \frac{Q_n \gamma_n (r - t \hat{x} \cdot \vec{B}_n)}{|\gamma_n^2 (t - r \hat{x} \cdot \vec{B}_n)^2 - t^2 + r^2|^{3/2}}$

* $F_{rt}(\vec{x}, t) = \frac{e^2}{4\pi} \sum_{n=1}^n \frac{Q_n \gamma_n (r - t \hat{x} \cdot \vec{B}_n)}{|\gamma_n^2 (t - r \hat{x} \cdot \vec{B}_n)^2 - t^2 + r^2|^{3/2}}$ *

$$r^2 = \vec{x} \cdot \vec{x}, \quad \vec{x} = r \hat{x}$$

$$\square A^\mu = e^2 j^\mu \quad (\text{Lorentz Gauge})$$

Coulomb Potential: $A_\mu(x) = \frac{e^2 Q U_\mu}{4\pi r}$

$\Rightarrow$ Decompose x $\perp$ and $\parallel$ to U^μ

$$x^\mu = x_\perp^\mu + a U^\mu$$

$$U^\mu x_\mu = a U^2 = -a, \quad a = -U^\mu x_\mu$$

$$\Rightarrow x^\mu = x_\perp^\mu - (U^\mu x_\mu) U^\mu$$

$$\begin{aligned} x^\mu x_\mu &= (x_\perp^\mu - (U \cdot x) U^\mu) (x_{\mu\perp} - (U \cdot x) U_\mu) \\ &= x_\perp^2 - 2(x_\perp U)(x_\perp U) - (U \cdot x)^2 \\ &= x_\perp^2 - (U \cdot x)^2 \end{aligned}$$

$\therefore \underline{x_\perp^2 = x^2 + (U \cdot x)^2}$, so ρ is the perp dist

• ρ spatial distance from the of $x \perp U^\mu$ worldline as measured in the instantaneous rest frame of Q_u

• Rest frame of $U^\mu = (1, \vec{0})$

$$U \cdot x = -t, \quad x^2 = -t^2 + r^2$$

$$\Rightarrow \rho^2 = x^2 + t^2 = r^2$$

In the frame of Q_u :

$$A_\mu(x) = \frac{e^2 Q}{4\pi r^2} \hat{t}_\mu$$

$$\hat{t}_\mu = (1, \vec{0})$$

$$\frac{dU^\mu}{d\tau} = 0$$

$$\partial_\mu A_\nu = \frac{e^2 Q}{4\pi r^2} U_\nu \partial_\mu \left(\frac{1}{\rho} \right) \Big|_{\rho^2 = x^2 + (U \cdot x)^2}$$

$$\begin{aligned} \partial_\mu (\rho^{-1}) &= -\frac{1}{\rho^2} \left(\frac{1}{2} \frac{1}{\rho} (2x_\mu + 2U_\mu (U \cdot x)) \right) \\ &= -\rho^{-3} (x_\mu + U_\mu (x \cdot U)) \end{aligned}$$

$$\Rightarrow \partial_\mu A_\nu(x) = -\frac{e^2 Q}{4\pi \rho^3} (x_\mu + U_\mu (x \cdot U)) U_\nu$$

$$F_{\mu\nu} = \partial_\mu A_\nu(x) - \partial_\nu A_\mu(x)$$

$$= \frac{e^2 Q}{4\pi \rho^3} (U_\mu x_\nu - U_\nu x_\mu)$$

$$= \frac{e^2 Q}{4\pi} \frac{(U_\mu x_\nu - U_\nu x_\mu)}{(x^2 + (U \cdot x)^2)^{3/2}}$$

$$\vec{B}_u = \frac{d\vec{x}}{dt}$$

$$U^\mu = \gamma_u (1, \vec{\beta}_u) \Rightarrow U_\mu = \gamma_u (-1, \vec{\beta}_u)$$

$$x^\mu = (t, \vec{x}) \rightarrow x_\mu = (-t, \vec{x})$$

$$U^\mu x_\mu = \gamma_k (-t + \vec{\beta}_k \cdot \vec{x}) \quad r \hat{x} = \vec{x}$$

$$U_\mu x = -\gamma_k t + \gamma_k r \vec{\beta}_k \cdot \hat{x}$$

$$(U_\mu x)^2 = \gamma_k^2 (t - r \hat{x} \cdot \vec{\beta}_k)^2, \quad x^2 = -t^2 + r^2$$

$$\Rightarrow \rho^2 = \gamma_k^2 (t - r \hat{x} \cdot \vec{\beta}_k)^2 - t^2 + r^2$$

$$\rho^3 = [\gamma_k^2 (t - r \hat{x} \cdot \vec{\beta}_k)^2 - t^2 + r^2]^{3/2}$$

$$\Rightarrow F_{\mu\nu} = \frac{e^2 Q}{4\pi} \left(\frac{U_\mu x_\nu - U_\nu x_\mu}{[\gamma_k^2 (t - r \hat{x} \cdot \vec{\beta}_k)^2 - t^2 + r^2]^{3/2}} \right)$$

$$\bullet \{ U_r = \hat{x}^i U_i = \gamma_k \hat{x} \cdot \vec{\beta}_k$$

$$\{ x_r = r = \hat{x}^i x_i$$

$$\{ U_t = -\gamma$$

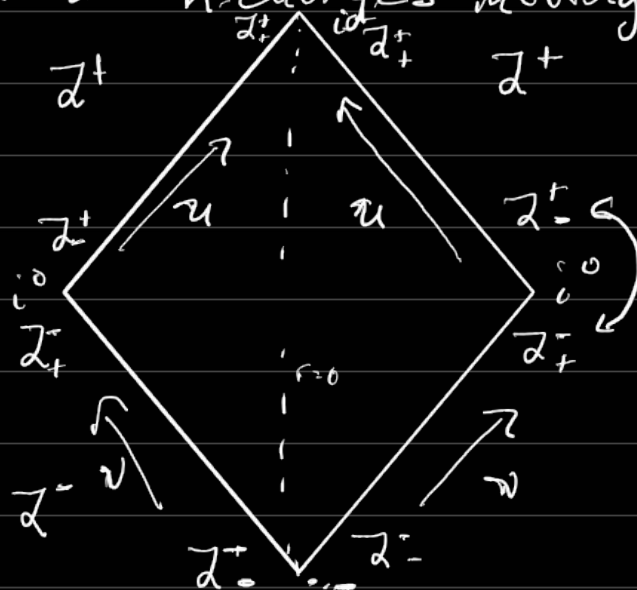
$$\{ x_t = -t$$

$$F_{rt} = \frac{e^2 Q}{4\pi} \left(\frac{\gamma_k \hat{x} \cdot \vec{\beta}_k (-t) - (-\gamma_k) r}{[\gamma_k^2 (t - r \hat{x} \cdot \vec{\beta}_k)^2 - t^2 + r^2]^{3/2}} \right)$$

$$\text{or } F_{rt} = \sum_{n=1}^N \frac{e^2 Q_n}{4\pi} \left[\frac{\gamma_n (r - t \hat{x} \cdot \vec{\beta}_n)}{[\gamma_n^2 (t - r \hat{x} \cdot \vec{\beta}_n)^2 - t^2 + r^2]^{3/2}} \right]$$

• Discontinuity at $r = \infty$, not single valued

* Want to look at n-charges moving at constant velocity



$\mathcal{I}_+^+$: Future of

* Note $\mathcal{I}_+^+ \in \mathcal{I}_+^-$
"connect" to $\mathcal{I}^0$
but $\neq \mathcal{I}^0$.

$$\mathcal{I}_\pm^\pm = S^2 \times L, \quad L: \text{null line}$$

* Scattering in M^D :

(i) Specify initial data on $\mathcal{I}^-$ and/or $\mathcal{I}^+$





(ii) Classically: find a classical map from the phase space defined on $\mathcal{I}^-$ to the phase space on $\mathcal{I}^+$

$$f: T^*\mathcal{I}^- \rightarrow T^*\mathcal{I}^+$$

Quantum: Find an S-matrix that evolves incoming states on $\mathcal{I}^-$ to outgoing on $\mathcal{I}^+$

$$\langle \text{out} | \hat{S} | \text{in} \rangle_{\mathcal{I}^\pm} = \langle \mathcal{O}_1(z, \bar{z}), \dots, \mathcal{O}_n(z, \bar{z}) \rangle$$

- What happens at i^0 is important as we need to match the data $\mathcal{I}_+^-$ to $\mathcal{I}_+^+$



- Might think to simply take each to be equal but this is wrong.

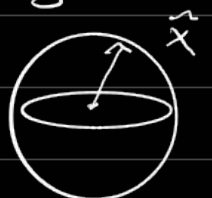
$$\left. \begin{array}{l} \mathcal{I}^-: v = t + r \quad (\text{advanced}) \\ \mathcal{I}^+: u = t - r \quad (\text{retarded}) \end{array} \right\} \text{3D spaces}$$

- Both t and r are infinite at $\mathcal{I}^\pm$

- $\hat{x}$: unit vector labeling a point on S^2

$$\mathcal{I}^+: (u, \hat{x})$$

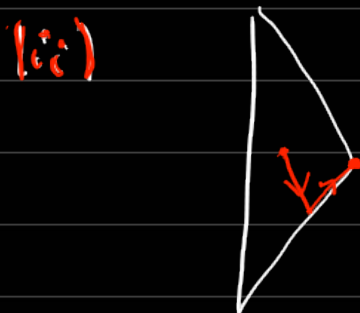
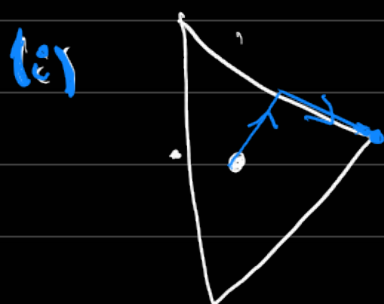
$$\mathcal{I}^-: (v, \hat{x})$$



Antipodal Matching:

Idea: (i) Start at a point in the Bulk and take paths to $\mathcal{I}^+$ and then to i^0 :

(ii) " $\mathcal{I}^- \rightarrow i^0$ "



(i) Rewrite F_{rt} in terms of $(u, \hat{x})$ and r

$$F_{rt}(\bar{x}, t) = \frac{e^2}{4\pi} \sum_{k=1}^n \frac{Q_k \gamma_k (r - t \hat{x} \cdot \vec{p}_k)}{|\gamma_k (t - r \hat{x} \cdot \vec{p}_k)^2 - t^2 + r^2|^{3/2}}$$

$$u = t - r \Rightarrow t = r + u$$

$$\Rightarrow F_{rt}(\bar{x}, t) = \frac{e^2}{4\pi} \sum_{k=1}^n \frac{Q_k \gamma_k (r - (u+r) \hat{x} \cdot \vec{p}_k)}{|\gamma_k^2 (u+r - r \hat{x} \cdot \vec{p}_k)^2 - (u+r)^2 + r^2|^{3/2}}$$

To reach $\mathcal{I}^+$: u : fixed, $r \rightarrow \infty$

$$F_{rt}(\bar{x}, t) \sim \frac{e^2}{4\pi} \sum_{k=1}^n \frac{r Q_k \gamma_k (1 - \hat{x} \cdot \vec{p}_k)}{r^3 |\gamma_k^2 (1 - \hat{x} \cdot \vec{p}_k)^2|^{3/2}}$$

$$F_{rt}(\bar{x}, t)|_{\mathcal{I}^+} \sim \frac{e^2}{4\pi r^2} \sum_{k=1}^n \frac{Q_k}{\gamma_k^2 (1 - \hat{x} \cdot \vec{p}_k)^2} \quad F_{rt}|_{\mathcal{I}^+}$$

getting to $\mathcal{I}^+$ means taking $u \rightarrow -\infty$ but since $F_{rt}(\bar{x}, t)|_{\mathcal{I}^+}$ does not depend on u

$F_{rt}(\bar{x}, t)$ on all of $\mathcal{I}^+$

* For a static set of charges $\vec{p}_k = 0$, $\gamma_k = 1$ leaving the usual Coulomb fields:

$$U_k = (1, \vec{0}) \quad \frac{e^2 Q}{r^2}, \quad Q = \sum_{k=1}^n Q_k$$

(Static): $F_{rt}|_{\mathcal{I}^+} = \frac{e^2 Q}{4\pi r^2}$

* Electrodynamics: look at multipole expansion of static configs. $\frac{1}{r^2}$ term: constant and proportional to the total electric charge

$\Rightarrow \vec{r}_3$: Comes from the static electric dipole moment with angular momentum $l=1$.

$\Rightarrow$ Moving charges: no dipole but $\exists$ dipole moment in the $\vec{r}_2$ term in the sense that the $l=1$ mode of the distribution over the sphere is nonzero.

* $\mathcal{I}^-$: $\mathcal{N} = t + r$

$$F_{rt} = F_{rv} = \frac{e^2}{4\pi} \sum_{k=1}^n \frac{Q_k \gamma_k (r - (v-r)\hat{x} \cdot \hat{B}_k)}{\gamma_k^2 ((v-r) - r\hat{x} \cdot \hat{B}_k)^2 - (v-r)^2 + r^2}^{3/2}$$

$$F_{rt}|_{\mathcal{I}^-} = \frac{e^2}{4\pi r^2} \sum_{k=1}^n \frac{Q_k}{\gamma_k^2 (1 + \hat{x} \cdot \hat{B}_k)^2}$$

• Since this is $\mathcal{N}$ independent ($v \rightarrow \infty$)

$$F_{rv}|_{\mathcal{I}^+} = \frac{e^2}{4\pi r^2} \sum_{k=1}^n \frac{Q_k}{\gamma_k^2 (1 + \hat{x} \cdot \hat{B}_k)^2}$$

• Field values do not match and is not equivalent at spatial infinity.

$\hookrightarrow$ Depends on how one approaches spatial ∞ and is discontinuous there. So long as $\vec{B}_k \neq 0$.

• Solution is Lorentz covariance: i.e. can be obtained by boosting a static charge. So, discontinuity is dictated by Lorentz invariance.

$\Rightarrow$ Vectors that generate Λ^μ_ν are singular at $i^\pm$

$$i^\pm \quad \Lambda^\mu_\nu = [\exp(UT)]^\mu_\nu \quad T|_{\mathcal{I}^\pm} : \text{singular}$$

• They have anti-podal matching via

$$\hat{x} \longleftrightarrow -\hat{x}$$

$$F_{rt}|_{\mathcal{I}^+} \quad F_{rt}|_{\mathcal{I}^-}$$

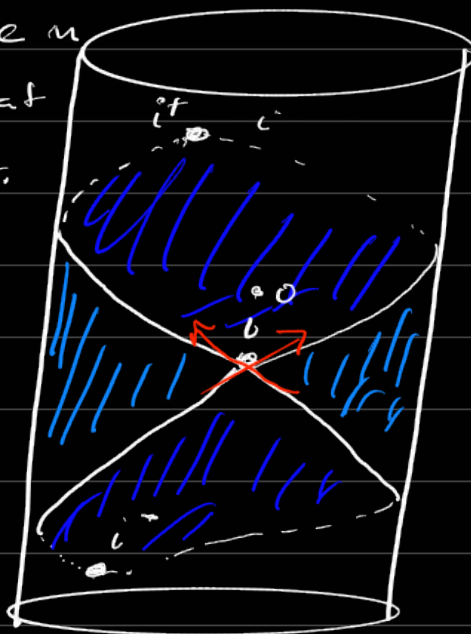
Matching condition: $\lim_{r \rightarrow \infty} r^2 F_{ru}(\hat{x})|_{\mathcal{I}^+} = \lim_{r \rightarrow \infty} r^2 F_{rv}(-\hat{x})|_{\mathcal{I}^-}$

• Since $F_{ru} = F_{rv} = F_{rt}$, note $\underline{F_E} = \underline{F_{E0}}$ so the above are values for the electric fields.

* One can check that the above holds in Lorentz frame

all reasonable circumstances of interactions.

- If we add electromagnetic waves, the field strengths at $\mathcal{I}^+$ and $\mathcal{I}^-$ would no longer be constant.
- Map $M_C^D \rightarrow S^3 \times \mathbb{R}$ (Einstein cylinder)
- By images, any single charge in the bulk will need an image at i^0 which will cause the elect. field to diverge there.



Asymptotic Expansions:

* Retarded coords $(u, z, \bar{z})$

$$ds^2 = -du^2 - 2du dr + 2r^2 \gamma_{z\bar{z}} dz d\bar{z}$$

• Penrose study:

$$\gamma_{z\bar{z}} = 2 / (1 + z\bar{z})^2$$

$$d\Omega_{(2)} = 2 \left(\frac{2}{(1+z\bar{z})^2} \right) dz d\bar{z}$$

$$= \left(\frac{2}{(1+z\bar{z})^2} \right) dz d\bar{z} + \left(\frac{2}{(1+z\bar{z})^2} \right) d\bar{z} dz$$

$$\Rightarrow \gamma = \begin{pmatrix} 0 & \gamma_{z\bar{z}} \\ \gamma_{\bar{z}z} & 0 \end{pmatrix}$$

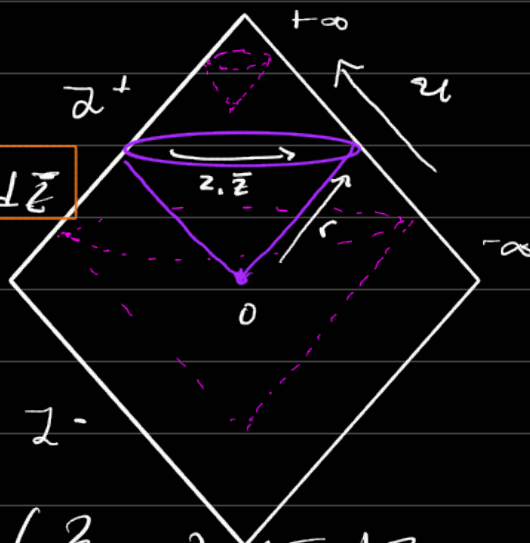
• Note that $\gamma^{z\bar{z}} \gamma_{z\bar{z}} = 2$

$$\Rightarrow \gamma^{z\bar{z}} = (1 + z\bar{z})^2$$

• On $\mathcal{I}^+$: $(u, z, \bar{z})$ constant, $r \rightarrow \infty$

$$\hookrightarrow du = dz = d\bar{z} = 0$$

leaves $ds^2 = 0$, so they are all indeed null lines
 $M^4 \rightarrow \text{Bordi/CRT}$



$$(x^0, \dots, x^3) \quad (u, r, z, \bar{z})$$

Coordinate transformations:

$$\frac{2rz}{1+z\bar{z}}$$

$$r^2 = (\vec{x})^2 \quad t = u + r, \quad x^1 + ix^2 = \frac{2rz}{1+z\bar{z}}$$

$$x^3 = \frac{r(1-z\bar{z})}{1+z\bar{z}}$$

inverse maps: $r^2 = (\vec{x})^2, \quad u = t - r$

$$z = \frac{x^1 + ix^2}{x^3 + r}$$

$\mathbb{R}^+$

z runs over the entire $\mathbb{C}$ -plane

$z=0$ north pole $x^1 = x^2 = 0$

$z=\infty$ South pole

$z\bar{z}=1$ equator, $z \rightarrow \frac{1}{z}$ antipodal map

$$\Rightarrow \mathcal{M}^4: ds^2 = -dt^2 + (d\vec{x})^2$$

$$= -dt^2 + dr^2 + r^2 d\Omega^2$$

$$d\Omega^2 = \frac{2dzd\bar{z}}{(1+z\bar{z})^2}$$

$$du = dt - dr \Rightarrow dt = du + dr$$

$$dt^2 = du^2 + 2drdu + dr^2$$

$$ds^2 = -du^2 - 2drdu + 2r^2 \gamma_{z\bar{z}} dzd\bar{z} \quad | \quad \mathbb{R}^+$$

• Constant values of

u means we have a 2-sphere from $u: -\infty \rightarrow \infty$ on $\mathbb{R}^+$



* On $\mathbb{R}^-$ we have $v = t + r$

$$ds^2 = -dv^2 + 2dvdr + 2r^2 \gamma_{z\bar{z}} dzd\bar{z}$$

$$\bar{x}^2 = r^2, \quad t = v - r, \quad x^1 + ix^2 = \frac{-2rz}{1+z\bar{z}}, \quad x^3 = -r \frac{1-z\bar{z}}{1+z\bar{z}}$$

$$z \rightarrow -\frac{1}{\bar{z}}$$

* Expand around z $A_z(u, r, z, \bar{z}) = \sum_{n=0}^{\infty} \frac{A_z^{(n)}(z, \bar{z})}{r^n}$ (11. denotes the order of the expansion)

about $r = \infty$

$$\Rightarrow F_{ru}^{(2)}(z, \bar{z})|_{\mathcal{I}^+} = F_{rv}^{(2)}(z, \bar{z})|_{\mathcal{I}^+}$$

- Evaluating at $\mathcal{I}^+$ means $u \rightarrow -\infty$

$$F_{ru}^{(2)}(z, \bar{z}) = F_{ru}^{(2)}(-\infty, z, \bar{z})$$

- Values of z on $\mathcal{I}^{\pm}$ are antipodal.

An ∞ of Conserved Charges: $\mathcal{E}: M^4 \rightarrow \mathbb{R}$ s.t.

$$B.C.: \mathcal{E}(z, \bar{z})|_{\mathcal{I}^+} = \mathcal{E}(z, \bar{z})|_{\mathcal{I}^-}$$

- $\mathcal{E}$ is not smooth near i^0 but instead is antipodally identified.

Define future and past charges:

$$Q_{\mathcal{E}}^+ := \frac{1}{e^2} \int_{\mathcal{I}^+} \mathcal{E} * F$$

$$Q_{\mathcal{E}}^- := \frac{1}{e^2} \int_{\mathcal{I}^-} \mathcal{E} * F$$

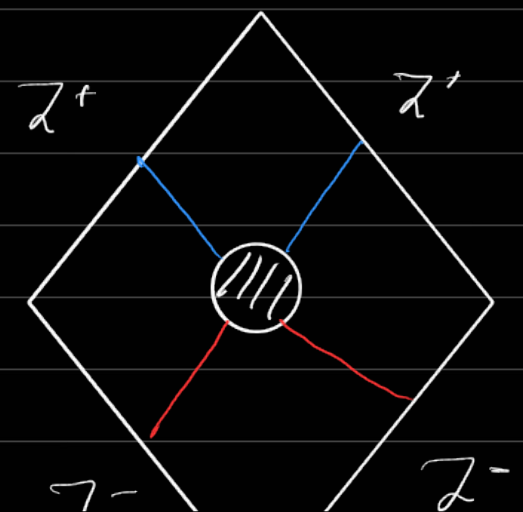
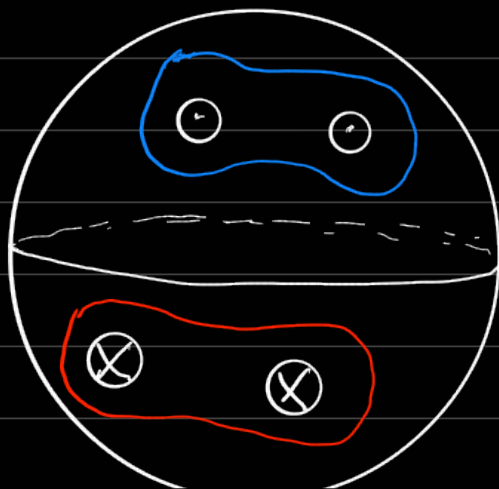
- Theory has an ∞ of conservation laws, one for every

$\mathcal{E}$: Can take expand $\mathcal{E}$ as $\mathcal{E}|_{\mathcal{I}^+} = \gamma_{em}$

From $F|_{\mathcal{I}^+} = F|_{\mathcal{I}^-}$ and $\mathcal{E}|_{\mathcal{I}^+} = \mathcal{E}|_{\mathcal{I}^-}$ we have that

$$Q_{\mathcal{E}}^+ = Q_{\mathcal{E}}^-$$

- All incoming multipole moments of the E.M. field are equal to the antipodal outgoing moments



- In the basis of $E|z_r = iem$, the antipodal map act as $(-)^l$.

$l=0$: total charge conservation:

$$E(z, \bar{z}) = \sum_{l=0}^{\infty} \sum_{m=-l}^l e_{lm} Y_{lm}(\theta, \phi)$$

- let $\Omega \equiv (z, \bar{z})$ on S^2 , then

$$E^+(\Omega) = E^-(P\Omega) = E^-(-\Omega)$$

P : antipodal map. $P\Omega = -\Omega$

* Note on asymptotic expansion: $A_2(u, r, z, \bar{z}) \sim \sum_{n=0}^{\infty} \frac{1}{r^n}$

$p \equiv \frac{1}{r}$, $dr = -dp/p^2$

$$ds^2 = \frac{1}{p^2} (-p^2 du^2 + 2 du dp + 2 \gamma_{z\bar{z}} dz d\bar{z})$$

$p=0 \Rightarrow r \rightarrow \infty$

$$A_2(u, p, \Omega) \sim A_2^{(0)}(u, \Omega) + p A_2^{(1)}(u, \Omega) + p^2 A_2^{(2)} + \dots$$

Replacing $p \rightarrow \frac{1}{r}$ yields

$$A_2(u, r, z, \bar{z}) \sim \sum_{n=0}^{\infty} \frac{A_2^{(n)}(u, z, \bar{z})}{r^n}$$

* $*F|_{S^2} = r^2 F_{\Omega\bar{\Omega}} d\Omega$

In retarded radial gauge: $A_r = 0$

$$\partial_e A_{\mu a} = \partial_{\mu} E \Rightarrow \partial_r E = 0$$

$$A_{\mu a} = A^{(0)} + A^{(1)} z_a + \dots$$

$$\partial A_{\mu a} = \partial_{z_a} E = 0$$

$$\Rightarrow A_{\mu a}^{(1)} = 0 \Rightarrow \partial A_{\mu a}^{(0)} = \partial_{z_a} E^{(0)} = 0$$

and so $E = E(z, \bar{z})$

- A scalar at i^0 must have a direction independent limit. $E(z, \bar{z})$ has different limits along angles so only $l=0$ part could be smooth at i^0 .

wd $Y_{lm}: S^2 \rightarrow \mathbb{C}$, form an orthon on $L^2_{\mathbb{C}}(S^2)$

$\Rightarrow f \in L^2_{\mathbb{C}}(S^2)$ must be square integrable

$$f(z, \bar{z}) = \sum_{l=0}^{\infty} \sum_{m=-l}^l f_{lm} Y_{lm}(z, \bar{z})$$

• Y_{lm} and Y_{lm}^* are orthonormal over the S^2 metric

$$\int d\Omega Y_{lm} Y_{l'm'}^* = \delta_{ll'} \delta_{mm'} \quad \left| \quad d\Omega = \sin\theta d\theta d\phi \right| \quad \text{with } r=1$$

• Or,
$$\sum_{l=0}^{\infty} \sum_{m=-l}^l Y_{lm}^* Y_{l'm'} = \delta(l-l') \delta(m-m') \quad \text{(Completeness relation)}$$

$$Q_E^+ = \frac{1}{e^2} \int_{\mathcal{I}^+} E(z, \bar{z}) * F = \frac{1}{e^2} \int d\Omega E(z, \bar{z}) F_{r\bar{r}}^{(2)}$$

$$E(z, \bar{z}) = \sum_{l=0}^{\infty} \sum_{m=-l}^l E_{lm} Y_{lm}(\theta, \phi)$$

* Note Q_E^+ is a map

$$Q_E^+ : C^\infty(S^2) \rightarrow \mathbb{R}$$

$$E \mapsto Q_E^+$$

Q_{lm}^+ : when $E \rightarrow Y_{lm}$

$$Q_{lm}^+ = \frac{1}{e^2} \int_{\mathcal{I}^+} Y_{lm}(z, \bar{z}) d\Omega F_{r\bar{r}}^{(2)}$$

• Note $Y_{lm}(-\Omega) = (-1)^l Y_{lm}(\Omega)$

$$Q_{lm}^+ = (-1)^l Q_{l,-m}^-$$

* Write Q as a surface integral and assume that all stable charged particles are massless

$$Q_E^+ = \frac{1}{e^2} \int_{\mathcal{I}^+} E * F \quad \left| \quad \begin{array}{l} \mathcal{I}^+ \cong \mathbb{R}_+ \times S^2 \\ \mathcal{I}^+ : \lim_{r \rightarrow +\infty} S_r^2 \quad \left| \quad \mathcal{I}^+ : \lim_{r \rightarrow \infty} S_r^2 \right. \\ \downarrow \quad \leftarrow i^* \quad \downarrow \end{array} \right.$$

• For an embedding $i : \mathcal{I}^+ \hookrightarrow \bar{M}$

$$\int_{\mathcal{I}^+} i^*(dE \wedge *F)$$

• Stokes' theorem: $N_{12}^+ = \{(z, \bar{z}) \in \mathcal{I}^+ \mid z_1 \leq z \leq z_2\}$
 $\cong [r_1, r_2] \times S^2$

2-form: $\omega := E * F$ on S^2

$$d\omega = d(E * F) = dE \wedge *F + E d * F$$

but Maxwell is $d * F = e^2 * j$

$$d\omega = d(E * F) = dE \wedge *F + E e^2 * j$$

$$\Rightarrow i^*(d\omega) = d_-(i^* \omega)$$

$$\partial N_{12}^+ = S^2|_{u_1} - S^2|_{u_2}$$

$$\Rightarrow \int_{N_{12}^+} d\omega = \int_{\partial N_{12}} \omega = \left(\int_{S_{u_1}^2} - \int_{S_{u_2}^2} \right) \omega$$

$$\bullet \int_{N_{12}^+} (dE \wedge *F + E e^2 *j) = \int_{S_1^2} E *u F - \int_{S_2^2} E *F$$

$$\lim_{\substack{u_1 \rightarrow -\infty \\ u_2 \rightarrow +\infty}} : \int_{\mathcal{I}_+^+} E *u F = Q_\Sigma^+ e^2 = \int_{\mathcal{I}_+^+} (dE \wedge *F + E e^2 *j) + \int_{\mathcal{I}_+^+} E *F$$

$$Q_\Sigma^+ = \frac{1}{e^2} \int_{\mathcal{I}_+^+} E *u F = \frac{1}{e^2} \int_{\mathcal{I}_+^+} dE \wedge *F + \int_{\mathcal{I}_+^+} E *j + \cancel{\frac{1}{e^2} \int_{\mathcal{I}_+^+} E *F}$$

$\Rightarrow$ Assumption: $\mathcal{I}$ charged massive particles

$$\bullet \frac{1}{e^2} \int_{\mathcal{I}_+^+} E *F = \frac{1}{e^2} \int d\Omega E F_{ru}^{(2)}(+\infty, \bar{z}, \bar{z})$$

$$\Rightarrow F_{ru}^{(2)}(+\infty, \bar{z}, \bar{z}) = 0$$

All massive particles must end at $\mathcal{I}^+$.

$$\bullet Q_\Sigma = \frac{1}{e^2} \int_{\mathcal{I}_+^+} dE \wedge *F + \int_{\mathcal{I}_+^+} E *j$$

$$\nearrow \Sigma|_{\mathcal{I}_+^+} = \Sigma|_{\mathcal{I}_+^-}$$

$\bullet$ Charge conservation: $Q_\Sigma^+ = Q_\Sigma^-$ however we restrict our case to

$$\partial_u \Sigma = \partial_v \Sigma = 0$$

$$\bullet \Sigma = \text{constant (i.e. } l=0)$$

$$Q_\Sigma = \Sigma \int *j$$

$\Rightarrow$ n : incoming

n : outgoing

$$Q_\Sigma^+ = Q_\Sigma^- \Leftrightarrow \sum_{n=1}^m Q_n^{\text{in}} = \sum_{k=1}^n Q_k^{\text{out}}$$

$\int E *j$: Sum over electric charges weighted by Σ
 $\frac{1}{e^2} \int dE \wedge *F$: Soft photon term

$\hookrightarrow$ when promoted to an operator it produces $a_{\vec{k},\alpha}^\dagger, a_{\vec{k},\alpha}$

zero energy photons.

• For retarded coordinates (Bondi) u :

$$Q_2^+ = \frac{1}{e^2} \int_{\mathcal{I}^+} \mathcal{E} * F = \frac{1}{e^2} \int d\Omega r^2 \mathcal{E}_{\theta\phi}{}_{\alpha\beta} F^{\alpha\beta}$$

$$= \frac{1}{e^2} \int \gamma_{z\bar{z}} dz d\bar{z} \mathcal{E} F_{ru}^{(2)}$$

$$Q_2^+ = \frac{1}{e^2} \int_{\mathcal{I}^+} d^2z \gamma_{z\bar{z}} F_{ru}^{(2)} \mathcal{E} = \frac{1}{e^2} \int_{\mathcal{I}^+} d^2z \gamma_{z\bar{z}} F_{ru}^{(2)} \mathcal{E} = Q_2^+$$

* Constraint Equation near $\mathcal{I}^+$.

$D^z, D_{\bar{z}}$: Covariant derivatives on S^2 with $\gamma_{z\bar{z}}$

$$D^z = \gamma^{z\bar{z}} D_{\bar{z}}$$

→ Leading Maxwell constraint:

$$\nabla_\mu F^{\mu\nu} = e^2 j^\nu$$

$\underbrace{N=r \text{ term}}: \nabla_\mu F^{\mu r} = e^2 j^r$

$$\nabla_\mu F^{\mu\nu} = \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} F^{\mu\nu})$$

$$\sqrt{g} = r^2 \sqrt{\gamma}, \quad \gamma \equiv \det(\gamma^{z\bar{z}})$$

$$\nabla_\mu F^{\mu r} = \frac{1}{r^2 \sqrt{\gamma}} \partial_\mu (r^2 \sqrt{\gamma} F^{\mu r}) = e^2 j^r$$

$$-r^2 \sqrt{\gamma} e^2 j^r = \partial_\mu (r^2 \sqrt{\gamma} F^{\mu r})$$

$$-r^2 \sqrt{\gamma} e^2 j^r = \partial_u F^{ur} r^2 \sqrt{\gamma} + r^2 \partial_2 (\sqrt{\gamma} F^{2r}) + r^2 \partial_{\bar{2}} (\sqrt{\gamma} F^{\bar{2}r})$$

* (u, r) -block

$$g_{ab} = \begin{pmatrix} g_{uu} & g_{ur} \\ g_{ru} & g_{rr} \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ -1 & 0 \end{pmatrix}$$

$$g^{ab} = \begin{pmatrix} g^{uu} & g^{ur} \\ g^{ru} & g^{rr} \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ -1 & 1 \end{pmatrix}$$

Since $g_{ab} g^{bc} = \begin{pmatrix} -1 & -1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \delta_a^c$

So $\left\{ \begin{matrix} g^{uu} = 0 \\ g^{ur} = -1 \end{matrix} \right\}, \left\{ \begin{matrix} g^{ru} = -1 \\ g^{rr} = 1 \end{matrix} \right\}$

$$j^r e^2 = \partial_u F^{ur} + \frac{1}{\sqrt{\gamma}} \partial_2 (\sqrt{\gamma} F^{2r}) + \frac{1}{\sqrt{\gamma}} \partial_{\bar{2}} (\sqrt{\gamma} F^{\bar{2}r})$$

$$e^2 (g^{ru} j_u + g^{rr} j_r) = \partial_u F^{ur} + \frac{1}{\sqrt{\gamma}} \partial_2 (\sqrt{\gamma} F^{2r}) + \frac{1}{\sqrt{\gamma}} \partial_{\bar{2}} (\sqrt{\gamma} F^{\bar{2}r})$$

$$\begin{aligned}
 F_{ur} &= g^{\mu\sigma} F_{\sigma\rho} g^{\rho r} \\
 &= (g^{\mu r}) (F_{ur} g^{rr}) \\
 &= -F_{ur} = F_{ru}
 \end{aligned}$$

$$\begin{aligned}
 F_{Ar} &= g^{A\sigma} F_{\sigma\rho} g^{\rho r} \\
 &= g^{AB} (F_{Bu} g^{ur} + F_{Br} g^{rr}) \\
 &= \frac{1}{r^2} \gamma^{AB} (-F_{Bu} + F_{Br})
 \end{aligned}$$

$\Rightarrow$

$$\begin{aligned}
 e^2(j_r - j_u) &= \partial_u(F_{ru}) + \frac{2}{\sqrt{\gamma}} \partial_A \left(\frac{1}{r^2} \gamma^{AB} (F_{Br} - F_{Bu}) \right) \\
 r^2 e^2(j_r - j_u) &= \partial_u(r^2 F_{ru}) + \frac{1}{\sqrt{\gamma}} \partial_A (\gamma^{AB} (F_{rB} - F_{rB}))
 \end{aligned}$$

• Define D_A on S^2 :

$$D_A V^A := \frac{1}{\sqrt{\gamma}} \partial_A (\sqrt{\gamma} \gamma^{AB} V_B)$$

$$r^2 e^2(j_r - j_u) = \partial_u(r^2 F_{ru}) + D^A (F_{uA} - F_{rB})$$

Radial Gauge: $A_r = 0$ & $A_u|_{\mathcal{I}^+} = 0$

$$A_u = \frac{A_u^{(1)}}{r} + \frac{A_u^{(2)}}{r^2}, \quad A_u^{(0)} = 0$$

$$A_A = A_A^{(0)} + \mathcal{O}(r^{-1})$$

$$F_{ru} = \partial_r A_u - \partial_u A_r$$

$$* F_{ru} = \frac{F_{ru}^{(2)}}{r^2} + \mathcal{O}(r^{-3})$$

$$\begin{aligned}
 * F_{uA} &= \partial_u A_A - \partial_A A_u \\
 &= F_{uA}^{(0)} + F_{uA}^{(1)}/r + \dots
 \end{aligned}$$

$$* F_{rA} = \partial_r A_A - \partial_A A_r = \partial_r A_A = \frac{F_{rA}^{(2)}}{r^2} + \mathcal{O}(r^{-3})$$

• Matter reaching $\mathcal{I}^+$ has

$$j_u = j_u^{(2)}/r^2 + \mathcal{O}(r^{-3}), \quad j_r = \mathcal{O}(r^{-4})$$

$\Rightarrow r^2 j_r$ must be finite

$$j^r = -j_u + \mathcal{O}(r^{-4})$$

$$r^2 \partial_u F_{ru} = r^2 \frac{\partial_u F_{ru}^{(2)}}{r^2} + \mathcal{O}(r^{-1})$$

$$D^A F_{uA} = D^A F_{uA}^{(0)} + \mathcal{O}\left(\frac{1}{r}\right)$$

$$D^A F_{rA} = (D^A F_{rA}^{(2)})/r^2 + \mathcal{O}(r^{-3}) \sim \mathcal{O}(r^{-2})$$

$$\Rightarrow$$

$$r^2 j_u \sim j_u^{(2)} + \mathcal{O}(r^{-1}), \quad r^2 j_r = \mathcal{O}(r^{-2})$$

Leading terms only means

$$-j_u^{(2)} e^2 = \partial_u F_{ru}^{(2)} + D^A F_{uA}^{(0)}$$

$$\therefore \boxed{e^{2+(2)} + \partial_u F_{ru}^{(2)} + D^A F_{uA}^{(0)}}$$

or expanded in $z, \bar{z}$:

$$\partial_u F_{ru}^{(2)} + D^z F_{uz}^{(0)} + D^{\bar{z}} F_{u\bar{z}}^{(0)} + e^{2+(2)}_{ju} = 0$$

(leading Maxwell term)

Choose $\partial_u z|_{\mathcal{I}^+} = 0$,

$$Q_z^+ = \frac{1}{e^2} \int_{\mathcal{I}^+} \mathcal{E} \star F = \frac{1}{e^2} \int_{\mathcal{I}^+} d^2 z \gamma_{z\bar{z}} \mathcal{E} F_{ru}^{(2)}$$

$$= \frac{1}{e^2} \int_{\mathcal{I}^+} d\mathcal{E} \wedge \star F + \int_{\mathcal{I}^+} \mathcal{E} \star j$$

$$d\mathcal{E} = \partial_z \mathcal{E} dz + \partial_{\bar{z}} \mathcal{E} d\bar{z} + \partial_u \mathcal{E} du$$

* If $\star F$ is over δ^2 then this is trivially

zero:

$$\begin{aligned} (\star F) &= \frac{1}{2} \frac{1}{\sqrt{g}} \epsilon_{\mu\nu\sigma\rho} F^{\sigma\rho} dx^\mu \wedge dx^\nu \\ &= \frac{1}{2} \mathcal{E} z u_{\sigma\rho} F^{\sigma\rho} dz \wedge du \\ &\quad + \frac{1}{2} \mathcal{E} \bar{z} u_{\sigma\rho} F^{\sigma\rho} d\bar{z} \wedge du \end{aligned}$$

$\Rightarrow$

$$\begin{aligned} d\mathcal{E} \wedge \star F &= \frac{1}{2} F^{\sigma\rho} \left[\partial_z \mathcal{E} \mathcal{E} \bar{z} u_{\sigma\rho} dz \wedge d\bar{z} \wedge du \right. \\ &\quad \left. + \partial_{\bar{z}} \mathcal{E} \mathcal{E} z u_{\sigma\rho} d\bar{z} \wedge dz \wedge du \right] \\ &= \left[\partial_z \mathcal{E} F^{\sigma\rho} \mathcal{E} u_{\sigma\rho} \bar{z} - F^{\sigma\rho} \partial_{\bar{z}} \mathcal{E} \mathcal{E} u_{\sigma\rho} z \right] \\ &\quad \times d\bar{z} \wedge dz \wedge du \end{aligned}$$

$$= (\partial_z \mathcal{E} F^{r\bar{z}} - F^{r\bar{z}} \partial_{\bar{z}} \mathcal{E})$$

$$\begin{aligned} \bullet F^{r\bar{z}} &= g^{r\mu} F_{\mu\nu} g^{\nu\bar{z}} = g^{r\mu} F_{\mu\bar{z}} \frac{1}{r^2} \gamma^{\bar{z}z} \\ &= F_{r\bar{z}} \frac{1}{r^2} \gamma^{\bar{z}z} \frac{1}{\sqrt{g}} + (-1) F_{u\bar{z}} \frac{1}{r^2} \gamma^{\bar{z}z} \frac{1}{\sqrt{r^2}} \end{aligned}$$

$\star AB$

$$\begin{pmatrix} 0 & 1 \\ r^2 & 0 \end{pmatrix} = -1$$

$$\Rightarrow (F_{r\bar{z}} \gamma^{2\bar{z}} - F_{u\bar{z}} \gamma^{2\bar{z}})$$

$$* \gamma^{2\bar{z}} \partial_z \epsilon (F_{r\bar{z}} - F_{u\bar{z}})$$

$$\bullet \partial_r A_{\bar{z}} - \partial_{\bar{z}} A_r = \partial_r A_{\bar{z}}$$

$$\bullet \text{Note } F_{rA} = -\frac{1}{r^2} A_A^{(1)} + \mathcal{O}(r^{-3}) = \mathcal{O}(r^{-2})$$

$$\text{So at } \mathcal{I}^+ \quad F_{rA}|_{\mathcal{I}^+} = 0, \text{ leaving } F_{u\bar{z}} \\ = -\gamma^{2\bar{z}} \partial_z \epsilon F_{u\bar{z}}$$

• Other term:

$$\begin{aligned} -\frac{1}{2} (\partial_{\bar{z}} \epsilon) F^{\sigma\rho} \epsilon_{\mu\sigma\nu\rho} &= -(\partial_{\bar{z}} \epsilon) F^{r\bar{z}} \\ &= (\partial_{\bar{z}} \epsilon) g^{r\mu} F_{\mu A} g^{\bar{z}A} = -(g^{r\mu} F_{uA} + g^{rr} F_{rA}) g^{\bar{z}A} \\ &= (F_{uA} - F_{rA}) g^{\bar{z}A} \partial_{\bar{z}} \epsilon \\ &= \partial_{\bar{z}} \epsilon \gamma^{2\bar{z}} (F_{u\bar{z}} - F_{r\bar{z}}) \\ &= (\partial_{\bar{z}} \epsilon) \gamma^{2\bar{z}} F_{u\bar{z}} \end{aligned}$$

$$\therefore \int_{\mathcal{I}^+} d\epsilon \wedge *F = \int_{\mathcal{I}^+} du d^2z \gamma^{2\bar{z}} (\partial_{\bar{z}} \epsilon F_{u\bar{z}} - \partial_z \epsilon F_{u\bar{z}})$$

• Missing a minus sign on first term and $\gamma^{2\bar{z}}$ is not present

$$\hookrightarrow - \int_{\mathcal{I}^+} du d^2z \gamma^{2\bar{z}} (\partial_{\bar{z}} \epsilon F_{u\bar{z}} + \partial_z \epsilon F_{u\bar{z}}) \gamma_{2\bar{z}}$$

$$\text{on } \mathcal{I}^+ \quad F_{uA} = F_{uA}^{(0)} + F_{uA}^{(1)}/r + \dots$$

$$\Rightarrow - \int_{\mathcal{I}^+} du d^2z \gamma^{2\bar{z}} (\partial_{\bar{z}} \epsilon F_{u\bar{z}}^{(0)} + \partial_z \epsilon F_{u\bar{z}}^{(0)}) \gamma_{2\bar{z}} \\ \text{on } \mathcal{I}^+ \text{ (Forget } \gamma_{2\bar{z}} \text{ in } \sqrt{-g})$$

* Charge conservation term

$$\int_{\mathcal{I}^+} \epsilon *j: \quad *j = \sqrt{-g} \epsilon_{\mu\nu\alpha\beta} g^{\sigma\rho} j_\rho dx^\mu dx^\nu dx^\alpha dx^\beta$$

$$\begin{aligned} (*j)_{u\bar{z}\bar{z}} &= \sqrt{-g} \epsilon_{u\bar{z}\bar{z}\sigma} g^{\sigma\rho} j_\rho \\ &= \sqrt{-g} \epsilon_{u\bar{z}\bar{z}\bar{z}} (g^{r\bar{z}} j_{\bar{z}} + g^{rr} j_r) \\ &= r^2 (-j_{\bar{z}} + j_r) \gamma_{2\bar{z}} \end{aligned}$$

$$\sim_{r \rightarrow \infty} r^2 (-j_{\bar{z}} + j_r) \sim -j_{\bar{z}}^{(2)} \gamma_{2\bar{z}}$$

$$(*j)_{u\bar{z}\bar{z}} = -j_{\bar{z}}^{(2)} \gamma_{2\bar{z}}$$

Again off by (1)

$$Q_\varepsilon^1 = \underbrace{-e^2 \int_{\mathbb{R}^4} d^4x (\partial_\mu \varepsilon F_{\mu\bar{z}}^{(0)} + \partial_{\bar{z}} \varepsilon F_{\mu z}^{(0)})}_{\equiv Q_S^+} + \underbrace{\int_{\mathbb{R}^4} d^4x \varepsilon \gamma_{z\bar{z}} j_\mu^{(2)}}_{\equiv Q_H^+}$$

Q_S^+ : "Soft" charge, linear in electromag. field

Q_H^+ : "Hard" charge, linear in charge current (massless).

• If $\varepsilon = \text{const.}$ then $Q_S = 0$

$$F_{\mu z}^{(0)} = \partial_\mu A_z^{(0)} - \partial_z A_\mu^{(0)} = \partial_\mu A_z(z, \bar{z})$$

$$\Rightarrow Q_S^+ = -\frac{1}{e^2} \int_{\mathbb{R}^4} d^4x (\partial_\mu \varepsilon F_{\mu\bar{z}}^{(0)} + \partial_{\bar{z}} \varepsilon F_{\mu z}^{(0)})$$

$$= -\frac{1}{e^2} \int d^2z \left[\partial_\mu \varepsilon \int_{-\infty}^{\infty} du F_{\mu\bar{z}}^{(0)} + \partial_{\bar{z}} \varepsilon \int_{-\infty}^{\infty} du F_{\mu z}^{(0)} \right]$$

Define: $\int_{-\infty}^{\infty} du F_{\mu z}^{(0)} \equiv N_z(z, \bar{z})$

• which is the $\omega \rightarrow 0$ limit of $\int_{-\infty}^{\infty} du F_{\mu z}^{(0)} e^{i\omega z}$ and the transverse to $\mathbb{R}^4$

• So, promoted to an operator

$$\hat{\mathcal{B}}(z, \bar{z}) \equiv \int_{-\infty}^{\infty} du \hat{F}_{\mu z}^{(0)} e^{i\omega u}$$

Creates and annihilates outgoing photons of energy ω

$$\Rightarrow Q_S^+ = -\frac{1}{e^2} \int d^2z (\partial_\mu \varepsilon N_{\mu\bar{z}}^{(0)} + \partial_{\bar{z}} \varepsilon N_{\mu z}^{(0)})$$

so $\partial_{\bar{z}} \varepsilon$ is the polarization vector

• Note

$$\partial_{\bar{z}} N_z - \partial_z N_{\bar{z}} = \int_{-\infty}^{\infty} du \left[\partial_{\bar{z}} F_{\mu z}^{(0)} - \partial_z F_{\mu\bar{z}}^{(0)} \right]$$

Bianchi: $dF = 0$

$$d\left(\frac{1}{2} F_{\mu\nu} dx^\mu \wedge dx^\nu\right) = \frac{1}{2} \partial_\sigma F_{\mu\nu} dx^\sigma \wedge dx^\mu \wedge dx^\nu$$

$$\Rightarrow \partial_{[\sigma} F_{\mu\nu]} = 0$$

$$(\partial_\sigma F_{\mu\nu} - \partial_\sigma F_{\nu\mu} + \partial_\nu F_{\sigma\mu} - \partial_\nu F_{\mu\sigma} + \partial_\mu F_{\nu\sigma} - \partial_\mu F_{\sigma\nu}) = 0$$

$$\therefore \left(\partial_\sigma F_{\mu\nu} + \partial_\nu F_{\sigma\mu} + \partial_\mu F_{\nu\sigma} \right) = 0$$

$$\partial_{\mu} F_{\bar{z}z}^{(\omega)} + \partial_{\bar{z}} F_{\bar{z}\mu}^{(\omega)} + \partial_{\mu} F_{\bar{z}z}^{(\omega)} = 0$$

$$\Rightarrow \partial_{\bar{z}} F_{\mu z}^{(\omega)} - \partial_z F_{\mu \bar{z}}^{(\omega)} = -\partial_{\mu} F_{\bar{z}z}^{(\omega)}$$

$$\partial_{\bar{z}} N_z - \partial_z N_{\bar{z}} = \int_{-\infty}^{\infty} du \left(-\partial_{\mu} F_{\bar{z}z}^{(\omega)} \right) = -F_{\bar{z}z}^{(\omega)} \Big|_{\mathcal{I}^-}^{\mathcal{I}^+}$$

• Assuming no magnetic monopoles and no long range magnetic fields $\Rightarrow F_{\bar{z}z}^{(\omega)}|_{\mathcal{I}^{\pm}} = 0$

$$\Rightarrow \partial_{\bar{z}} N_z = \partial_z N_{\bar{z}} \quad (\text{Kahler Cauchy})$$

$$\hookrightarrow \boxed{N_z \equiv e^2 \partial_z N} \quad \left| \begin{array}{l} \text{Gauge } A_{\mu}^{(0)} = 0 \\ \text{choose } \partial_{\mu} \varepsilon \text{ s.t.} \end{array} \right.$$

$$N_z = e^2 \partial_z N = \int_{-\infty}^{\infty} du F_{\bar{z}z}^{(\omega)} = \int_{-\infty}^{\infty} du \partial_{\mu} A_z^{(\omega)} = A_z^{(\omega)} \Big|_{\mathcal{I}^+} - A_z^{(\omega)} \Big|_{\mathcal{I}^-}$$

$$\boxed{N_z = A_z^{(\omega)} \Big|_{\mathcal{I}^+} - A_z^{(\omega)} \Big|_{\mathcal{I}^-}}$$

• To have finite energy the gauge field must be pure gauge on $\mathcal{I}^{\pm}$

$$\hookrightarrow \text{Gauge shift: } e^2 N = f$$

$$A_z \rightarrow A_z + \partial_z e^2 N$$

Soft photon charge:

$$\boxed{Q_{\varepsilon}^+ = 2 \int d^2 z N \partial_z \partial_{\bar{z}} \varepsilon + \int du d^2 z \gamma_{z\bar{z}} \hat{J}_{\mu}^{(2)}}$$

• Since

$$Q_{\varepsilon}^+ = -\frac{1}{e^2} \left(\int d^2 z (\partial_z \varepsilon e^2 \partial_{\bar{z}} \varepsilon N + e^2 \partial_{\bar{z}} \varepsilon \partial_z N) + \dots \right)$$

$$= - \left(\int d^2 z [\partial_z \varepsilon N] \Big|_{\mathcal{I}^+} + \int d^2 z [\partial_{\bar{z}} \varepsilon N] \Big|_{\mathcal{I}^-} + \dots \right)$$

$\hookrightarrow$ First two terms vanish, leaving the above

Note Q_{ε}^+ is an ∞ of conserved charges since ε can be ∞ many different spherical harmonics.

• Identified as a Dirac Bracket of charges on phase space

Canonical Electrodynamics at $\mathcal{I}^{\pm}$:

$$\bullet \Pi \equiv T^*U \sim v \cdot n \sim \{a_i, p_i\} \quad i=1,2 \quad N/2$$

- In field theory i_{ij} become continuous I and Γ becomes infinite dimensional.

Poincare 1-form: $\lambda = p_i dq^i$

$$\Rightarrow d\lambda = dp_i \wedge dq^i \in \omega^2$$

Def: A 2-form ω^2 on an even dim mfd M^{2n} is called symplectic provided ($M \rightarrow \text{symp. mfd}$)

(i) $d\omega = 0$

(ii) ω is nondegenerate: the linear map assoc. to a vector X , the 1-form

$i_X \omega^2$ is non singular

$$i_X \omega^2 \neq 0$$

$$[i_X \omega]_i = X^j \omega_{ji} \neq 0 \Rightarrow \det(\omega_{ij}) \neq 0$$

$\omega^n \equiv$ Liouville/Symplectic volume of Γ

* $L = L(q, \dot{q})$, on TM , then $P: TM \rightarrow T^*M$

$$P(q, \dot{q}) = \left(q, \frac{\partial L}{\partial \dot{q}^i} \right)$$

If H is time dependent then $\Gamma = T^*M \times \mathbb{R}$

$$\frac{dH}{dt} = \frac{\partial H}{\partial q^i} \frac{dq^i}{dt} + \frac{\partial H}{\partial p_i} \frac{dp_i}{dt} + \frac{\partial H}{\partial t} = \frac{\partial H}{\partial t}$$

* "Def" space is the space of fields, δ acts as d on this ∞ , dim space $(x, \phi(x))$

Euler-Lagrange den

$$\bullet \Omega_\Sigma = -\frac{1}{e^2} \int_\Sigma \delta(*F) \wedge \delta A$$

$\hookrightarrow$ (3)-form on Σ

$\int$ (2)-form on $\Gamma \leftarrow P$

* For Ω_Σ to be invertible, it must have zero modes

$\Rightarrow \delta A$ is restricted by gauge conditions, constraints δ, B, C .

$$\frac{1}{e^2} \int d\Sigma^\mu \delta F_{\mu\nu} \wedge d\Sigma^\nu$$

$$\Omega_\Sigma = e^2 \int_\Sigma$$

Take $\Sigma = \mathcal{I}^+$: $d\Sigma^\mu \rightarrow du d^2z$

$$\Omega_{\mathcal{I}^+} = \frac{1}{e^2} \int du d^2z \left(\delta F_{uz}^{(0)} \wedge \delta A_z^{(0)} + \delta F_{u\bar{z}}^{(0)} \wedge \delta A_{\bar{z}}^{(0)} \right)$$

• Separate out $A_z^{(0)}(u)$ - component.

$$A_z^{(0)}(u, z, \bar{z}) = \hat{A}_z(u, z, \bar{z}) + \partial_z \phi(z, \bar{z})$$

$$\partial_z \phi \equiv \frac{1}{2} \left[A_z^{(0)}|_{\mathcal{I}^+} + A_{\bar{z}}^{(0)}|_{\mathcal{I}^+} \right]$$

$$\Omega_{\mathcal{I}^+} = \frac{1}{e^2} \int du d^2z \left[\delta F_{uz}^{(0)} \wedge \delta \hat{A}_z + \delta F_{u\bar{z}}^{(0)} \wedge \delta \hat{A}_{\bar{z}} + \delta F_{u\bar{z}}^{(0)} \wedge \partial_{\bar{z}} \phi \right]$$

• Recall $N_z \equiv \int_{-\infty}^{\infty} du F_{uz}^{(0)} = e^2 \partial_z N = A_z^{(0)}|_{\mathcal{I}^+} - A_{\bar{z}}^{(0)}|_{\mathcal{I}^+}$

$$\delta F_{uz}^{(0)} = \delta \partial_u (\hat{A}_z) \text{ and } \delta F_{u\bar{z}}^{(0)} = \delta \partial_u \hat{A}_{\bar{z}}$$

$$\Omega_{\mathcal{I}^+} = \frac{1}{e^2} \int du d^2z \left[\partial_u \delta \hat{A}_z \wedge \delta \hat{A}_{\bar{z}} + \partial_u \delta \hat{A}_z \wedge \partial_{\bar{z}} \phi + \partial_u \delta \hat{A}_{\bar{z}} \wedge \delta \hat{A}_z + \partial_u \delta \hat{A}_{\bar{z}} \wedge \partial_z \phi \right]$$

$$\begin{aligned} \Omega_{\mathcal{I}^+} &= \frac{2}{e^2} \int du d^2z \left(\delta F_{uz}^{(0)} \wedge \delta A_z^{(0)} + \delta F_{u\bar{z}}^{(0)} \wedge \delta A_{\bar{z}}^{(0)} \right) \\ &= \frac{1}{e^2} \int du d^2z \left[\partial_u (\delta \hat{A}_z) \wedge \delta \hat{A}_{\bar{z}} + \partial_u \delta \hat{A}_z \wedge \delta \partial_{\bar{z}} \phi \right. \\ &\quad \left. + \partial_u \delta \hat{A}_{\bar{z}} \wedge \delta \hat{A}_z + \partial_u \delta \hat{A}_{\bar{z}} \wedge \delta \partial_z \phi \right] \\ &= \frac{2}{e^2} \int du d^2z \left[\partial_u (\delta \hat{A}_z) \wedge \delta \hat{A}_{\bar{z}} + \frac{1}{2} \int d^2z \left\{ \delta \left(\int du F_{uz}^{(0)} \right) \wedge \delta \partial_{\bar{z}} \phi \right. \right. \\ &\quad \left. \left. + \left(\delta \int du F_{u\bar{z}}^{(0)} \right) \wedge \delta \partial_z \phi \right\} \right] \\ &= \frac{2}{e^2} \int du d^2z \left[\partial_u (\delta \hat{A}_z) \wedge \delta \hat{A}_{\bar{z}} + \int d^2z \left\{ \delta (e^2 \partial_z N) \wedge \delta \partial_{\bar{z}} \phi \right. \right. \\ &\quad \left. \left. + \delta (e^2 \partial_{\bar{z}} N) \wedge \delta \partial_z \phi \right\} \right] \end{aligned}$$

$$\Omega_{\mathcal{I}^+} = \frac{2}{e^2} \int du d^2z \left[\partial_u (\delta \hat{A}_z) \wedge \delta \hat{A}_{\bar{z}} - 2 \int d^2z \partial_z \delta \phi \wedge \partial_{\bar{z}} \delta N \right]$$

Soft photon mode N_z | ϕ : Sum of Boundary values

• Radiative components of $A_z^{(0)}$ are paired

• Soft photon modes are paired with ϕ

* Idea: $\delta \mathcal{L} = E(d) \wedge \delta \phi + d \Theta$

$$\Omega = \int_\Sigma \delta \Theta \sim \int_\Sigma \delta p^i \wedge \delta q^i$$

and Ω allows us to read off the canonical variables: Here

there are two

Q: How does this apply to a solution space of the EOM?

Commutators:

$$[\hat{A}_z(z, \bar{z}), \hat{A}_{\bar{w}}(w', \bar{w})] = \frac{-ie^2}{2} \delta(z-w') \delta^2(\bar{z}-\bar{w})$$

Not an integral over w : this would yield boundary terms

$$\delta(z-w') = \partial_z \Theta(z-w') = \frac{1}{i\pi} \int dw e^{i\omega(z-w')}$$

$$\Rightarrow \Theta(z-w') = \frac{1}{i\pi} \int dw \frac{1}{\omega} e^{i\omega(z-w')}$$

$$[\hat{A}_z(z, \bar{z}), \hat{A}_{\bar{w}}(w', \bar{w})] = -\frac{ie^2}{4} \Theta(z-w') \delta^2(\bar{z}-\bar{w})$$

$$[\partial_z \phi(z, \bar{z}), \partial_{\bar{w}} N(w, \bar{w})] = \frac{i}{2} \delta^2(z-w)$$

$$\bullet \text{ Aside on: } \bar{\partial} \bar{\partial} \ln|z-w|^2 = 2\pi \delta^2(z-w)$$

$$\hookrightarrow \bar{\partial} \bar{\partial} (\ln(z-w) + \ln(\bar{z}-\bar{w}))$$

$$= \bar{\partial} \left(\frac{1}{z-w} \right)$$

$$\hookrightarrow \partial_{\bar{z}} \left(\frac{1}{z-w} \right) = 2\pi \delta^2(z-w) \quad \leftarrow \text{Distributional equality}$$

$$\iint f(z, \bar{z}) \partial_{\bar{z}} \left(\frac{1}{z-w} \right) d^2z = - \iint \partial_{\bar{z}} f \left(\frac{1}{z-w} \right) d^2z + (i) \oint_{\partial D} \frac{f}{z-w} dz$$

$$= 2\pi i f(w, \bar{w}) = 2\pi \int d^2z \delta^2(z-w) f(z, \bar{z})$$

$$\Rightarrow \partial_{\bar{z}} \left(\frac{1}{z-w} \right) = 2\pi \delta^2(z-w)$$

$$\bullet \text{ Regulator: } \ln(|z|^2) \rightarrow \ln(|z|^2 + \epsilon)$$

$$\partial \bar{\partial} \ln(|z|^2 + \epsilon) = \partial \left(\frac{z}{|z|^2 + \epsilon} \right) = \bar{\partial} \left(\frac{\bar{z}}{|z|^2 + \epsilon} \right)$$

$$= \left(\frac{1}{|z|^2 + \epsilon} \right) + \frac{(-1) z \bar{z}}{(|z|^2 + \epsilon)^2}$$

$$= \epsilon / (|z|^2 + \epsilon)^2$$

$$\Rightarrow \int d^2z \frac{\epsilon}{(|z|^2 + \epsilon)^2} f(z, \bar{z}) \rightarrow \text{go to polar } (z, \bar{z}) \rightarrow (r, \theta)$$

$$|z|^2 = r^2 = \rho$$

$$\int_0^\infty d\rho \int d\theta \frac{\epsilon}{(\rho + \epsilon)^2} f(\rho, \theta) = \int_0^\infty d\rho \frac{\epsilon}{(\rho + \epsilon)^2} g(\rho)$$

$$= \left(-\frac{\epsilon}{\rho + \epsilon} g(\rho) + \epsilon \ln(\rho + \epsilon) g'(\rho) \right) \Big|_0^\infty - \int_0^\infty d\rho \epsilon \ln(\rho + \epsilon) g''(\rho)$$

$$= g(0) = 2\pi f(0)$$

$$\Rightarrow -\partial_z \partial_{\bar{w}} \ln|z-w|^2 = 2\pi \delta^2(z-w)$$

$$[\partial_z \phi, \partial_{\bar{w}} N] = \frac{i}{2} \delta^2(z-w) = -\frac{i}{4\pi} \partial_z \partial_{\bar{z}} \ln|z-w|^2$$

or

$$[\phi(z, \bar{z}), N(w, \bar{w})] = -\frac{i}{4\pi} \ln|z-w|^2 + f(z, \bar{z}) + g(w, \bar{w})$$

Large Gauge Symmetries:

$$[Q_z^+, A_z^{(0)}(z, \bar{z})] = i \partial_z \mathcal{E}(z, \bar{z}) \stackrel{?}{=} i \delta_z A_z^{(0)}$$

Proof:

$$\left[2 \int d^2z N \partial_z \partial_{\bar{z}} \mathcal{E} + \int_{\mathcal{S}^1} d\tau d^2z \mathcal{E} \gamma_{z\bar{z}} j_z^{(2)}, A_z^{(0)}(z, \bar{z}) \right]$$

$\Rightarrow$ there $[j_z^{(2)}, A_z^{(0)}] = 0$?

$$e^2 j_z = \nabla^\mu F_{\mu z} = D^z F_{z z} + D^{\bar{z}} F_{\bar{z} z}$$

$$= -D^z (\partial_z A_z^{(0)}) - D^{\bar{z}} (\partial_z A_{\bar{z}}^{(0)})$$

$$= -D^z (\partial_z \hat{A}_z) - D^{\bar{z}} \partial_z \hat{A}_{\bar{z}}$$

$$- \frac{i}{2} \left(D^z \left[-\frac{e^2}{2} \delta(z-w) \delta^2(z-w) \right] + D^{\bar{z}} (\dots) \right)$$

$$\frac{i}{2} \delta(z-w) \left[D^z \delta^2(z-w) + D^{\bar{z}} \delta^2(z-w) \right] + \partial_z F_{z z}$$

$$\left(\frac{i}{2} \delta(z-w) \right) \delta^2(z-w) = \delta^2(z-w)$$

$$\left(\frac{i}{2} \delta(z-w) \right) (\gamma^{\bar{z} z} \partial_{\bar{z}} \delta^2(z-w) + \gamma^{\bar{z} z} \partial_z \delta^2(z-w))$$

= 0

$\Rightarrow$ leaves only first term:

$$\left[2 \int d^2z N \partial_z \partial_{\bar{z}} \mathcal{E}, A_z^{(0)}(z, \bar{z}) \right]$$

$$= 2 \left[\int d^2z N \partial \bar{\partial} \mathcal{E}, \partial \phi \right] = 2 \int d^2z \partial \bar{\partial} \mathcal{E} [N, \partial \phi]$$

$$= -2 \int d^2z \partial \mathcal{E} [\bar{\partial} N, \partial \phi] = 2 \int d^2z \partial \mathcal{E} \frac{i}{2} \delta^2(z-w)$$

$$= \partial_z \mathcal{E}(z, \bar{z}) \quad \checkmark \quad \text{as claimed.}$$

* Also find that: $[Q_z^-, A_z^{(0)}(z, \bar{z})] = i \partial_z \mathcal{E}(z, \bar{z})$

Trivially:

$$[Q_z^+, N(z, \bar{z})] = 0$$

$$[Q_z^+, \hat{A}_z(z, \bar{z})] = 0$$

$$[Q_z^+, \phi(z, \bar{z})] = i \mathcal{E}(z, \bar{z})$$

Maxwell: ∞ -# of symmetries generated by conserved charges

$Q_\epsilon^\pm$ in a canonical formalism are just gauge trans. with parameter ϵ

- For constant ϵ (i.e. γ_{00}), $Q_{\epsilon=1}^+$
 $\delta A = 0$
- For arbitrary ϵ (higher harmonics) $Q_\epsilon^\pm$ transforms A_z by a "large" gauge transformation which is non-zero at $\mathcal{I}^\pm$.
 $\lim_{r \rightarrow \infty} \epsilon \rightarrow \epsilon(\theta, \phi) = \epsilon(z, \bar{z})$ (angle dependent)
- $A_z = 0$ is not invariant under these symmetries
 $\Rightarrow$ they are spontaneously broken
 $\delta A_z = \delta A_z^{(0)} + \dots$
 $= i \partial_z \epsilon$ is not in general zero (only for constant ϵ)

* Now suppose we add charged matter

$\Rightarrow$ Above unchanged so long as the coupling approaches a constant in the IR

ex: $\mathcal{L} = -\frac{F^2}{4} \int F \wedge * F + \int d^4x \nabla_\mu \Phi^\dagger \nabla^\mu \Phi$
 $\nabla_\mu \sim \partial_\mu - i A_\mu q^2$

$\Rightarrow$ coupling term $\pm i e^2 A^\mu (\Phi \partial_\mu \Phi^\dagger - \Phi^\dagger \partial_\mu \Phi)$

Retarded radial Gauge: $A_r = 0$

$$A_z|_{\mathcal{I}^\pm} = 0$$

$$i e^2 A^\mu (\Phi \partial_\mu \Phi^\dagger - \Phi^\dagger \partial_\mu \Phi)$$

$\Rightarrow$ For massless charged matter which reaches $\mathcal{I}^\pm$, the energy flux through $\mathcal{S}^2$

$$F_\Phi = \lim_{r \rightarrow \infty} \int d\Omega r^2 T_{\mu\nu}^\Phi$$

$$T_{\mu\nu}^\Phi \sim (D_\mu \Phi)^\dagger (D_\nu \Phi)$$

Supposing the $\Phi \sim r^{-p} \Phi^{(p)}(u, z, \bar{z})$

$$\Rightarrow D_\mu \Phi = r^{-p} \partial_\mu \Phi^{(p)}(u, z, \bar{z})$$

$$F_\Phi = \lim_{r \rightarrow \infty} \int d\Omega r^2 (r^{-p})^2 (\partial_\mu \Phi^{(p)})^\dagger (\partial_\nu \Phi^{(p)})$$

$$p \geq 0 \quad 2 - 2p$$

if $p < 1$, this diverges, so

$$p_{\text{min}} = 1 \quad (\infty, p = 0)$$

$$r \rightarrow \infty \quad \begin{cases} \text{finite} & p=2 \\ 0 & p \geq 1 \end{cases}$$

So only $\Phi^{(1)}$ contributes at $\mathcal{I}^+$, $\Phi^{(0)} = 0$
to remove all unphysical terms.

$$\Phi = \frac{\Phi^{(1)}(u, z, \bar{z})}{r} + \mathcal{O}(r^{-2})$$

Now above, we showed $[Q_{\text{soft}}, A] \sim \text{large}$

$$\hookrightarrow Q_{\mathcal{E}}^+ = Q_S^+ + Q_H^+$$

$$[Q_{\mathcal{E}}^+, A] = 0$$

With matter added $\Omega = \begin{pmatrix} \Omega_M & 0 \\ 0 & \Omega_{EM} \end{pmatrix}$
and must check $Q_{\mathcal{E}}^+$

properly generates gauge transformations of Φ .

$\hookrightarrow$ Role of the hard charge

global
gen: Conserved Noether current of the $U(1)$
(canonically generates this symm on $\mathcal{I}^+$)

$$[Q_{\mathcal{E}}^+, \Phi_k(u, z, \bar{z})] = \left[\int_{\mathcal{I}^+} d^2u d^2\omega \mathcal{E} \gamma_{\omega\bar{\omega}} j_u^{(2)}, \Phi_k(u, z, \bar{z}) \right]$$

$$= \int d^2u d^2\omega \mathcal{E}(\omega, \bar{\omega}) \gamma_{\omega\bar{\omega}} [j_u^{(2)}(u', \omega, \bar{\omega}), \Phi_k(u, z, \bar{z})]$$

$$= i \delta_{\mathcal{E}} \Phi_k(u, z, \bar{z})$$

Note that

$$\boxed{\delta_{\mathcal{E}} \Phi_k(u, z, \bar{z}) = i Q_k \mathcal{E}(z, \bar{z}) \Phi_k(u, z, \bar{z})} \quad \text{✗}$$

$$\hookrightarrow e^{iQ_{\mathcal{E}} \Phi}$$

so

$$\int d^2u d^2\omega \mathcal{E}(\omega, \bar{\omega}) \gamma_{\omega\bar{\omega}} [\cdot, \cdot] = -Q_k \mathcal{E}(z, \bar{z}) \Phi_k(z, \bar{z}, u)$$

$$= \int d^2u' d^2\omega \delta(u - u') \delta^2(z - \omega) \gamma^{\omega\bar{\omega}} \gamma_{\omega\bar{\omega}} (-Q_k \mathcal{E}(\omega, \bar{\omega}) \Phi_k(z, \bar{z}, u))$$

$$\boxed{[j_u^{(2)}(u', \omega, \bar{\omega}), \Phi_k(u, z, \bar{z})] = -Q_k \gamma^{z\bar{z}} \Phi_k(u, z, \bar{z}) \delta(u - u') \delta^2(z - \omega)}$$

and

$$\boxed{[Q_H^+, \Phi_k(u, z, \bar{z})] = i \delta_{\mathcal{E}} \Phi_k(u, z, \bar{z})}$$

where $Q_H^+ = \int d^2x$

* The sum of Q_A, Q_S properly generates local angle dependent gauge transformations on $\mathcal{Z}^+$

Ward Identity:

Conserved charges commute with the Hamiltonian and hence the S-matrix. $S \sim \exp(iHT)$, $T \rightarrow \infty$

$\langle \text{out} | S | \text{in} \rangle$

$\Rightarrow$ Charge conservation:

$$\langle \text{out} | Q_\varepsilon^+ S - S Q_\varepsilon^- | \text{in} \rangle = 0$$

$\hookrightarrow$ Matching conditions: $Q_\varepsilon^+ = Q_\varepsilon^-$ | From $F_{\mu\nu}^{(2)}|_{\mathcal{Z}^+} = F_{\mu\nu}^{(2)}|_{\mathcal{Z}^-}$

$$\Rightarrow \vec{Q}_\varepsilon = 0 = [Q, H]$$

$$\langle \text{out} | [Q, H] | \text{in} \rangle = \langle \text{out} | (QH - HQ) | \text{in} \rangle$$

Q_ε^- acts on in states

Q_ε^+ acts on out states

$$\hookrightarrow \langle \text{out} | Q_\varepsilon^+ H - H Q_\varepsilon^- | \text{in} \rangle = 0$$

• For some field with $e^{iQ_\varepsilon \alpha} \Phi$

$$[e^{iQ_\varepsilon \alpha}, H] = 0$$

$$\langle \text{out} | (e^{iQ_\varepsilon^+ \alpha} H - H e^{iQ_\varepsilon^- \alpha}) | \text{in} \rangle = 0$$

Claim: A large Gauge transformation on the in-state evolves into a large Gauge transformation of out-state

$$Q_\varepsilon^- = 2 \int d^2z N \partial \bar{\partial} \varepsilon + \int_{\mathcal{Z}^-} d^2z \varepsilon \gamma_{z\bar{z}} j_z^{(2)}$$

$$Q_\varepsilon^- | \text{in} \rangle = \left(-2 \int d^2z \partial N \partial \varepsilon + \int_{\mathcal{Z}^-} d^2z \varepsilon \gamma_{z\bar{z}} j_z^{(2)} \right) | \text{in} \rangle$$

$$| \text{in} \rangle \sim \Phi_{i_1} \dots \Phi_{i_n} | 0 \rangle$$

$$\Rightarrow j_z^{(2)} \Phi_{i_1} = \Phi_{i_1} j_z^{(2)} + (-Q_{i_1}) \gamma^{z\bar{z}} \partial^2(z-w) \delta(z-z_1) \Phi_{i_1}$$

$$\therefore j_z^{(2)} \left(\prod_{k=1}^n \Phi_{i_k} \right) = \left(\prod_{k=1}^n \Phi_{i_k} \right) j_z^{(2)} - \sum_{k=1}^n Q_{i_k} \gamma^{z\bar{z}} \partial^2(z-w) \delta(z-z_k) \left(\prod_{k=1}^n \Phi_{i_k} \right)$$

$j_z^{(2)} | \text{in} \rangle = \dots$ the second term becomes

$$\int d^2z d^2\bar{z} \mathcal{E}(z, \bar{z}) \sum_{n=1}^{\infty} \dots |in\rangle$$

$$= - \sum_{n=1}^{\infty} Q_{in} \mathcal{E}(z_n^{in}, \bar{z}_n^{in}) |in\rangle$$

Act with $\Phi^\dagger \Rightarrow ([j, \Phi])^\dagger = \dots$

$$[j, \Phi^\dagger] = +(\dots)$$

and so

$$Q_{\bar{z}} |in\rangle = \underbrace{\left(-2 \int d^2z \partial_z N^{(c)} \partial_{\bar{z}} \mathcal{E} \right)}_{\text{Soft charge}} + \underbrace{\sum_{n=1}^m Q_n^{in} \mathcal{E}(z_n^{in}, \bar{z}_n^{in})}_{\text{Hard-charge}} |in\rangle$$

$$\langle out | Q_{\bar{z}}^\dagger = 2 \int d^2z \partial_z \partial_{\bar{z}} \mathcal{E} \langle out | N(z, \bar{z}) + \sum_{n=1}^m Q_n^{out} \mathcal{E}(z_n^{out}, \bar{z}_n^{out}) \langle out |$$

$$2 \int d^2z (\partial_z \partial_{\bar{z}} \mathcal{E}) \langle out | N(z, \bar{z}) S - S N(z, \bar{z}) | in \rangle$$

$$= \sum_{n=1}^m \left[Q_n^{in} \mathcal{E}(z_n^{in}, \bar{z}_n^{in}) - Q_n^{out} \mathcal{E}(z_n^{out}, \bar{z}_n^{out}) \right] \langle out | S | in \rangle$$

(Word Identity)'s

as ∞ # of word: are for each $\mathcal{E} \in \mathcal{F}_{em}(z, \bar{z})$

Soft factor := $Q_n^{in} \mathcal{E}(z_n^{in}, \bar{z}_n^{in}) - Q_n^{out} \mathcal{E}(z_n^{out}, \bar{z}_n^{out})$

Take the case $\mathcal{E}(w, \bar{w}) = \frac{1}{z-w}, \quad \partial_{\bar{z}} \left(\frac{1}{z-w} \right) = 2\pi \delta^{(2)}(z-w)$

LHS:

$$-4\pi \langle out | \partial_z N S - S N^{(c)}(z, \bar{z}) | in \rangle$$

$$= \sum_{n=1}^m \left(Q_n^{in} \frac{1}{z-z_n^{in}} - Q_n^{out} \frac{1}{z-z_n^{out}} \right) \langle out | S | in \rangle$$

$\Rightarrow$ Large Symmetry group is equivalent to a DDLac Modly symm. on $S^2|_{\mathbb{R}^+}$

Mode Expansion:

Expand A_2 as a plane wave solution to make contact with QFT.

* Both use the symplectic form

$$\Omega_{\Sigma} = -\frac{i}{2} \int_{\Sigma} \delta(*F) \wedge \delta A$$

→ Above we have $\Sigma = \mathbb{R}^+$

→ QFT $\Sigma = M_{t=0}$ (4) Plane-wave basis

$$ds^2 = -dt^2 + d\vec{x} \cdot d\vec{x}$$

Near $\mathbb{R}^+$, A_ν has an expansion

$$A_\mu(x) = e \sum_{\alpha=\pm} \int \frac{d^3q}{(2\pi)} \left(\frac{1}{2\omega} \right) \left[\epsilon_\mu^\alpha(\vec{q}) a_\alpha(\vec{q}) e^{iqx} + \epsilon_\mu^\alpha(\vec{q}) a_\alpha(\vec{q})^\dagger e^{-iqx} \right]$$

$$q^2 = 0,$$

Polarization vectors $\epsilon_\alpha^\mu \epsilon_\beta^{\mu*} = \delta_{\alpha\beta}$

$$[a_\alpha^{\text{out}}(\vec{q}), a_\beta^{\text{out}}(\vec{q}')^\dagger] = \delta_{\alpha\beta} (2\pi)^3 (2\omega_q) \delta^{(3)}(\vec{q} - \vec{q}')$$

• Next step, rewrite quantities on $\mathbb{R}^+$ in terms of $a, a^\dagger$

$$\hookrightarrow z = t - r, \quad ds^2 = -dz^2 - 2dzdr + 2\gamma_{z\bar{z}} r^2 dz d\bar{z}$$

$$\vec{x}^2 = r^2, \quad x^1 + ix^2 = \frac{z\bar{z}}{1+z\bar{z}}, \quad x^3 = r \frac{1-z\bar{z}}{1+z\bar{z}}$$

$$z = \frac{x^1 + ix^2}{x^3 + r}$$

$$* q^0 = \omega = r$$

$$q^\mu = (q^0, q^1, q^2, q^3) = (\omega, q^1, q^2, q^3)$$

$$\vec{q} = \left(\omega, r \frac{(z+\bar{z})}{(1+z\bar{z})}, \frac{-i(z-\bar{z})r}{(1+z\bar{z})}, \frac{-r(1-z\bar{z})}{(1+z\bar{z})} \right)$$

$$= \frac{\omega}{(1+z\bar{z})} (1+z\bar{z}, (z+\bar{z}), -i(z-\bar{z}), (1-z\bar{z}))$$

$z=0$: North pole $x^3 = r$

$z \rightarrow \infty$: South Pole

→ ex: at $z=0$, $q^\mu = \omega(1, 0, 0, 1)$

exactly a null vector in the t^3 -direction

• Choose ϵ_μ to be orthogonal to q^μ

→ Can have $\epsilon_\mu = (0, \alpha, \beta, 0)$ at $z=0$

$$\Rightarrow \alpha\alpha^* + \beta\beta^* = 1$$

$$\epsilon^\mu q_\mu = 0 = -(1+z\bar{z})\alpha + (z+\bar{z})\beta - i(z-\bar{z})\delta + (1-z\bar{z})\gamma = 0$$

$$= -\alpha - (z\bar{z})(\alpha+r) + z(\beta-i\delta) + \bar{z}(\beta+i\delta) + \gamma = 0$$

$$\alpha = -\gamma$$

$$\beta = \gamma$$

$$\boxed{\beta = -\alpha},$$

$$\boxed{\alpha = -\beta}$$

$$\xi^\mu = (\alpha, i\delta, \delta, -\alpha)$$

$$\rightarrow 2\alpha = 2i\delta\bar{z} \Rightarrow \boxed{\alpha = i\delta\bar{z}}$$

$$\xi^\mu = \delta(i\bar{z}, i, 1, -i\bar{z})$$

$$= i\delta(\bar{z}, 1, -i, -\bar{z})$$

$$\xi^\mu \xi_\mu^* = \delta^2(-2\bar{z} + 1 + 1 + 2\bar{z}) = 1$$

$$\Rightarrow \delta^2(2) = 1 \Rightarrow \boxed{\delta = \frac{1}{\sqrt{2}}}$$

which gives us exactly the polarization veds stated:

$$\xi^{+\mu}(\bar{q}) = \frac{1}{\sqrt{2}}(\bar{z}, 1, -i, -\bar{z}), \quad \xi^{-\mu}(q) = \frac{1}{\sqrt{2}}(z, 1, i, -z)$$

$$q^\mu \xi_\mu^{\pm}(\bar{q}) = 0,$$

$$\xi_a^\mu \xi_{B\mu}^* = \delta_{aB}$$

• Consider $\mathcal{A}^+$ field $A_2^{(0)}(z, \bar{z}, \bar{z})$

$$A_2^{(0)}(z, \bar{z}, \bar{z}) = \lim_{r \rightarrow \infty} A_2(r, z, \bar{z}, \bar{z})$$

$$A_2 = e \int \frac{d^2 q}{(2\pi)^2} \frac{1}{2\omega} \left[\xi_z^{*+} \alpha_+^{\text{out}} e^{iqx} + \xi_z^+ \alpha_+^{\text{out}} e^{-iqx} \right] + (-1)$$

$$\xi_z^+ = \int \frac{z A}{2\omega^2} \xi^{A+} = \xi^{\bar{z}+} \int \frac{z \bar{z}}{2\omega^2} = \omega^2 \frac{z}{(1+z\bar{z})^2} \xi^{\bar{z}+}$$

$$= \frac{\sqrt{2}\omega^2}{(1+z\bar{z})^2} \left(\frac{1}{\sqrt{2}} \right) (-\bar{z})$$

$$\xi_z^+ = \frac{-\sqrt{2}\omega^2}{(1+z\bar{z})^2} \bar{z}, \quad (\xi_z^+)^* = \frac{-\sqrt{2}\omega^2}{(1+z\bar{z})^2} z$$

$$\xi_{\bar{z}}^- = \frac{-\sqrt{2}\omega^2}{(1+z\bar{z})^2} z, \quad \hat{x} = \begin{pmatrix} \frac{z+\bar{z}}{1+z\bar{z}} & \frac{-i(z-\bar{z})}{1+z\bar{z}} & \frac{1-2\bar{z}}{1+z\bar{z}} \end{pmatrix}$$

Note that $\hat{x}$ is a unit vector which points to $(z, \bar{z})$ on S^2

is a unit vector which points to $(z, \bar{z})$ on S^2

